

Healthcare Applications of Respiratory Inductance Plethysmography: A Narrative Review

Abstract. Accurate interpretation of respiratory functions enables for early diagnosis of life-threatening disorders such as asthma, pneumonia, chronic obstructive pulmonary disease (COPD), tuberculosis (TB), and others, as well as providing useful evidence - based guidance. With the advent of technology, a broad selection of breathing monitoring devices are now accessible. Respiratory Inductance Plethysmography (RIP) is one of these instruments which allows non-invasive technique to acquire respiratory patterns and signals. The goal of this study is to present a detailed overview of recent advancements in RIP devices, as well as to highlight major applications and methodologies in present era and to appraise future research directions. The paper collects databases from IEEE, Science Direct, PubMed, and Google Scholar – were surveyed. The papers that demonstrated the major enabling characteristics of healthcare applications were included. The study found that RIP is a useful and accurate tool for assessing and monitoring respiratory function.

Keywords: Respiratory Inductance Plethysmography (RIP), COPD, wearable sensors, lung volume, non-invasive

1 Introduction

Accurate assessment of breathing/respiratory function permits early detection of potential respiratory disorders and provides insightful information for treatment. Over the few decades, several studies on breath monitoring systems and pulmonary function testing using sensors have been investigated. Pulmonary function tests (PFT) which primarily are vital assessments in the care of patients with respiratory illness, whether suspected or diagnosed. In medical applications, the pneumotachometer (PnT) is considered as the gold standard for determining tidal volume (TD) change; nevertheless, the use of a PnT requires the individual to wear a mouthpiece, indicating the change in endogenous ventilation and numerous PFT methods, however, such as spirometry and plethysmography with body box measurement, are time-consuming and need participant cooperation [1], [2].

Every year, significant resources are empowered to facilitate medical diagnosis, develop wearable sensors and create technology to identify various diseases. The application of wearable sensor technology assisting health-monitoring domain has also been significantly increased. Recently, breathing monitoring devices are getting a subject of interest in health profession and research community. Accurate assessment of breathing/respiratory function permits early detection of potential respiratory disorders and provides insightful information for treatment. Over the few decades, several studies on breath monitoring systems and pulmonary function testing using sensors

have been investigated. Pulmonary function tests (PFT) which primarily are vital assessments in the care of patients with respiratory illness, whether suspected or diagnosed. They can assist with diagnosis, monitoring therapy response, and guiding preventive and therapeutic recommendations. Significant efforts have been made in developing devices to monitor PFT and other respiration assessment including chest wall motion, airflow and respiratory volumes, respiratory rate, resulting transcutaneous oxygen saturation (SpO_2), and respiratory CO_2 removal. Table 1 classifies devices and methods used for respiration assessment.

Table 1. Categories of Sensing Principle for Respiratory Monitoring Devices and Methods

Measured Technique	Measured Category	Measured Parameters	Sensor Position
RIP, EIP ¹ , Full Body Plethysmography, Spirometry	Lung Movement, Volume & Tissue Composition	ABD ² & THC ³ Circumference change, FEV1 ⁴ , FVC ⁵ , FRC ⁶ , TLC ⁷ , RV ⁸	ABD and chest wall
ABG ⁹ Test	Arterial gas concentration	pH, P_{aO_2} ¹⁰ , P_{aCO_2} ¹¹ , SBC_e ¹² , HCO_3^- ¹³ , $(tCO_2(P))_e$ ¹⁴	Blood gas measurement Arterial gas concentration Peripheral organ or Nasal/oral area

Apart from the traditional techniques of respiratory parameter measurement, respiratory inductive plethysmography (RIP), is a noninvasive measurement of lung ventilation which consists of two elastic belts one in the abdomen around the umbilicus and one in the rib cage around the nipple line, to assess spontaneous tidal breathing by the movement of the chest during respiration. Generally, RIP can be done on subjects in around 5-10 mins [3]. Conventionally the RIP bands are incorporated with transducer essentially with coils that are connected to an analogue circuit which is a variable frequency LC oscillator. The coils act as an inductive element and therefore the frequency and amplitude can be calculated to find the inductance. The length of the

¹ Electrical Impedance Plethysmography

² Abdomen

³ Thoracic

⁴ Forced Expiratory Volume in 1 second

⁵ Forced Vital Capacity

⁶ Functional Residual Capacity

⁷ Total Lung Capacity

⁸ Residual Volume

⁹ Arterial Blood Gas

¹⁰ Arterial O_2 Partial Pressure

¹¹ Arterial CO_2 Partial Pressure

¹² Concentration of bicarbonate

¹³ Metabolic acidosis/alkalosis

¹⁴ Total CO_2

belt's changes during breathing, and the self-inductance of the coils changes proportionately. The variations can be identified in the circumference or cross-sectional area of the rib cage and belly by monitoring changes in the coils' self-inductance. The basic function of RIP is shown in Figure 1.

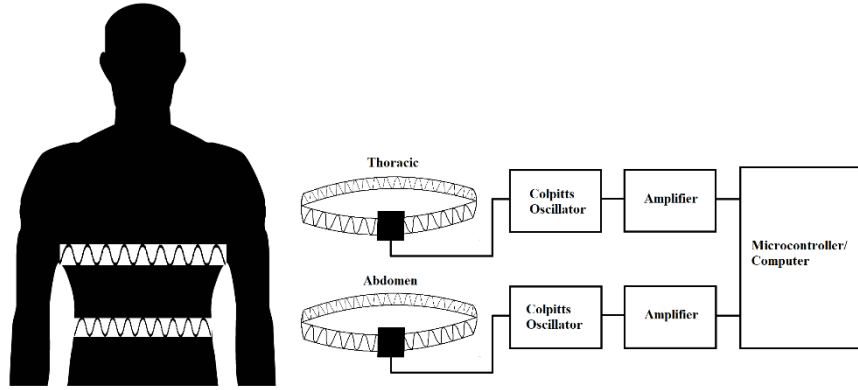


Fig. 1. Shows a basic block diagram for RIP

While, most of the existing technologies show promise, the major advantage of RIP over PMT, is that the tidal breathing in patients who are unable to cooperate due to their old age or disease severity, can also be examined through easy set of equipment [4]. On top that, the RIP system can also be implemented for individuals in an ambulatory environment [5,6]. The ease of this system, the device and method are becoming a big interest to researchers and clinicians. Unlike a detailed review on RIP devices, the aim of this work is to provide a short review to the reader about the key enabling features of RIP device, potential applications in health sector through disease detection, design highlights in various settings and purposes.

2 Key Enabling Features

2.1 Advancement of RIP

Over the time, the RIP device evolved depending on the applications. One important aspect of the RIP is that it is an oscillator-based hardware which requires calibration over a period of time. Additionally, it also requires several inductive sensors to obtain mutual inductance between RIP sensors. To address this issue with RIP, Cohen's et al in [7]; Zhang et al. [8] have suggested that the frequencies of the thoracic and abdominal band oscillators must be suitably spaced in attempt to lessen modulation and avoid frequency locking. Typically, a respiration tracking system comes with several inductive sensors, generating a steady oscillator for each inductive sensor while preserving frequency separation is complicated. For a battery-driven portable device,

energy consumption is critical and hence a small printed circuit board size becomes essential for mobility. Therefore, if multiple inductive sensors are involved, a traditional RIP design employing an LC oscillator increases chip count and power management intricacy [9]. To mitigate this problem, Sinton [10] devised a RIP system that uses a traditional Electrical Impedance Plethysmograph (EIP) to quantify the coil's self-inductance, since a steady current travels through the belts, this approach does not require an LC oscillator. Sinton's method incorporates a continuous AC signal which although resolves the frequency locking but requires high power which makes the system impossible to be portable. Zhang et al. [10] enact a Pulse Amplitude Modulation (PAM) technique for indirect inductance measurement without the need of a variable-frequency oscillator [9, 11, 12]. During the pulse excitation phase, a strong current from a constant-current source is momentarily and routinely given to the RIP sensor, and its inductance is measured. Unlike Sinton's method, the feeding current does not pass through the inductive sensor continually. As a result of the large feeding current at the time, the PAM method significantly decreases power usage while preserving sensitivity. Contemporary developments have completely diminished the use of an LC oscillatory circuit by using a high-performance, low-power embedded microprocessor which instantly digitizes the inductance, therefore, requiring no calibration and allowing to store data locally and simultaneously transmitting it wirelessly over computer and cloud using *pneuRIP*TM [13]. Therefore, advisable to the user to account the advancement in RIP device before applying to desired task.

2.2 Wearability and Mobility

Continuous monitoring of respiration is crucial for detecting and anticipating high-risk situations where an appropriate monitoring technique may pave the way to life-saving conditions [14]. The implementation of RIP in a wearable form factor allows to monitor PFT and BP of the user in a completely unrestrained environment, therefore allowing to measure data while the individual can carry on their daily activities. The mobility of RIP is also advantageous in ambulatory situations, where PFT and BP are critical to measure. Having such a system embedded in an ambulance might help diagnose patients, and provide critical reports for medics.

Wearable shirts with embedded sensors for monitoring vital indicators, including blood pressure, are available from Carre Technologies in Canada. The Hexoskin Smart Shirt (HX) was created with activity sensors in mind for usage in elite athletics and therapeutic settings [15]. This sort of wearable shirt can also be highly beneficial in other disciplines of research, such as cigarette smoking monitoring [16].

3 Applications of RIP Sensor in Disease Recognition

From a clinical standpoint, 'respiratory activity' provides aspects such as respiratory rate and depth, as well as quantitative information on the degree of gas exchange that is taking place. Continuous respiratory activity monitoring is crucial for recognizing or anticipating high-risk diseases, and correct monitoring techniques may save a person's life.

3.1 Apnea and Hypopnea

Obstructive sleep apnea-hypopnea syndrome (OSAHS) is a serious medical disorder that has become more common in recent decades. It is a major source of illness and death in both developed and developing countries [17].

For the identification of respiratory events during sleep studies, the American Academy of Sleep Medicine (AASM) advises using an oronasal thermistor and a nasal cannula [18]; nevertheless, these sensors are unpleasant for some individuals and can even disrupt their sleep. They can be inaccurate since patients can move them around or even remove them during sleep examinations. Sabil, A et al. [19] suggested that the combination of tracheal sounds and respiratory inductance plethysmography could be used as a reliable method for the detection and characterization of respiratory events without the need to place sensors on the face of patients and could be used either as an alternative or in combination with the recommended American Academy of Sleep Medicine sensors.

When given to young babies, anesthesia increases the risk of apnea in the postoperative period, a condition known as postoperative apnea (POA) [20, 21]. There have been reports of both central and obstructive apnea [22]. The absence of inspiratory effort characterizes central apneas. Due to the obstruction in the pharyngeal airway, obstructive apneas induce respiratory effort but no inspiratory outflow. Both central and obstructive apnea is life-threatening since the majority of newborns require medical intervention to re-establish breathing; fatalities have been documented [23].

An Automated Unsupervised Respiratory Event Analysis (AUREA) to identify breathing patterns acquired with dual belt RIP was implemented by [24]. The results given in this paper show that AUREA may be used to swiftly categorize breathing patterns in signals obtained from RIP, sample by sample. The AUREA pattern classifications have also been used to evaluate preterm neonates' breathing patterns after the endotracheal tube (ETT) has been withdrawn. To construct a predictor of extubation readiness in the extreme neonatal newborn, both AUREA patterns and AUREA metrics will be employed. As a result, analyzing the patterns and metrics produced from AUREA may indicate breathing trends and predict sensitive infants with the potential for POA.

3.2 Neuromuscular Diseases

Neuromuscular diseases are a wide term that refers to a variety of disorders that affect muscle function, either directly via pathologies of the voluntary muscle or indirectly through pathologies of the peripheral nervous system or neuromuscular junctions. Respiratory failure in neuromuscular illness might manifest as an abrupt occurrence or a chronic, progressive disease [25]. Approaches to measuring breathing in neuromuscular illnesses can be costly, time-consuming, and stressful for patients. Better assessment procedures, such as in-home examinations, are being explored. The pneuRIP might be a more efficient way to assess a patient's lung health. In [26] an outpatient clinic environment was set up and evaluates the efficacy of the pneuRIP in adolescents with neuromuscular illness and evaluate their RIP parameters to those of age-matched healthy individuals. The pneuRIP technique successfully evaluated the Work of Breathing (WOB) indices and respiratory patterns and has proven to be effective in recognizing abnormalities in WOB indices and breathing patterns in patients with neuromuscular illness.

3.3 Other Restrictive and Obstructive Lung Diseases

In medical practice, body plethysmography can deduce both restrictive and obstructive lung diseases such as Asbestosis, Pulmonary fibrosis, Sarcoidosis, Scoliosis, Asthma, Bronchiectasis, and Emphysema Cystic fibrosis, and Chronic bronchitis. Along with these, other cardiovascular diseases can also be monitored from the PFT using RIP. Therefore, the RIPs can be calibrated and/or designed for the identification of a specific disease in a noninvasive manner and a varied set of ambulatory environments.

4 Design and Performance of RIP

The design and performance of RIP has been vigorously tested over the years. Since the RIP operates based on belts, the design of wire woven or sewed along the belt, as well as its specifications, influence the RIP sensor qualities. The work in [27] studied the influence of step size of sine wave pattern on RIP sensor characteristics, and built three RIP sensors with the same wire pattern (sine wave), height (2.2 cm), and length (30 cm) but various step sizes, as shown in Fig. 2. The inductance of each of these belts were measured using Agilent 4285A Precision LCR Meter. Sensitivity and linearity of sensors with different step sizes are depicted in Fig. 3. It is evident from the Fig. 3 that the smaller step size, the higher the sensitivity. In term of linearity, there exists slight difference between 1 cm and 1.5 cm step size. However, for 3 cm of step size, linearity decreases remarkably. It is therefore advisable to keep the step size smaller to achieve good linearity.

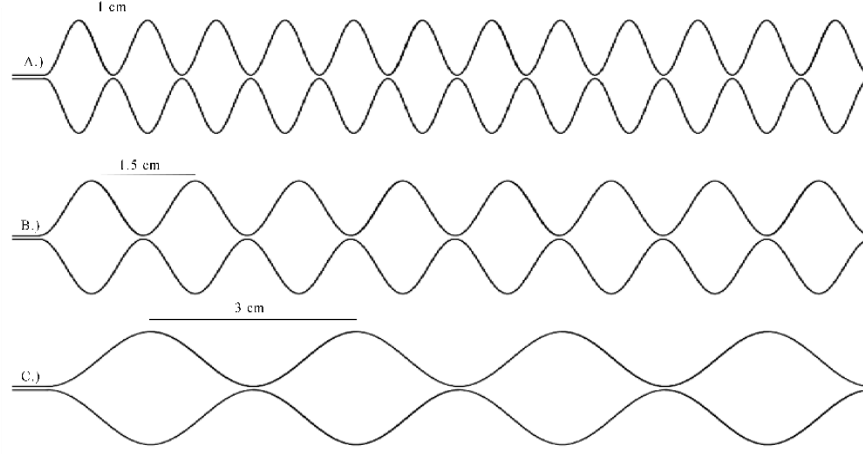


Fig. 2. RIP belt coil design with three different sine wave patterns with step sizes of A.) 1 cm, B.) 1.5cm C.) 3cm [26]

The researchers also contributed to investigate the placement of RIP sensors. The authors in [13] scrutinized three different attachment options for the RIP, (i) RIP belt secured using tension only; (ii) RIP belt held in place by adhesive and tension; (iii) RIP belt sewn into a T-Shirt. From the findings over the course of 24 hours, the method with the RIP belt stitched inside a T-Shirt had the highest performance. This approach has been proven to be more resistant to abnormalities induced by body movement. The RIP was only to be removed during showering, and participants in the trial were instructed to go about their normal everyday routines. The calibration coefficients of the system based merely on the belt or belt+adhesive become unreliable over time. This happened as a result of their day-to-day activities, making them unreliable and inconvenient. Table II demonstrate the system's average overall volume and standard deviation, whilst Table III reveals the data as calculated Root Mean Square Error (RMSE).

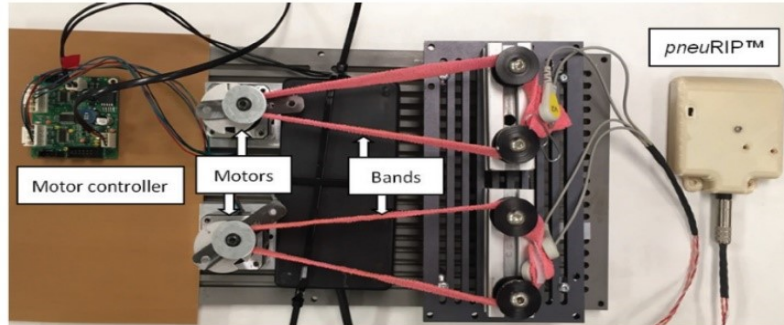
Table 2. System Comparison in terms of Volume in Liters (L) [13]

System	Average Volume (L)		
	Subject 1	Subject 2	Subject 3
Belt	0.577±0.093	0.999±0.212	1.094±0.295
Belt+Adhesive	0.761±0.445	1.295±0.272	0.994±0.517
Belt+Shirt	0.831±0.111	0.735±0.116	0.798±0.127

Table 3. System Comparison in terms of RMS in Liters (L) [13]

System	Average Volume (RMS)		
	Subject 1	Subject 2	Subject 3
Belt	0.223	0.199	0.294
Belt+Adhesive	0.039	0.495	0.194
Belt+Shirt	0.031	0.066	0.002

In [19], the researchers have conducted a Benchtop testing on the *pneuRIP*TM to determine the circuitry and instrument design, reliability, responsiveness, and repeatability. The RIP bands were mounted on a laboratory-grade rocker/waver platform as shown in Fig. 4. To mimic different breathing situations, the tilt angles and speed were changed. Over the course of two days, the testing ran without taking any pauses. The benchtop testing was performed along with RespiTrace system (Sensormedics) a RIP system from 1991 [13]. At typical breathing frequencies and phase angles, both systems (*pneuRIP*TM and RespiTrace) produce similar results but the phase angle variance was substantially reduced using *pneuRIP*TM for low and high phase angles, as well as low and high breathing frequency. *pneuRIP*TM is more precise and reliable at greater and lower phase angles because mechanical fluctuation is less. Children with respiratory distress frequently have large phase angles, which *pneuRIP*TM is intended to detect. A study [27] suggested that the step size should be smaller to achieve good linearity. Regarding the placement of the device, it was discouraged to use the RIP device with belt and adhesive

**Fig. 3.** Benchtop test-bed that simulates calibrated breathing patterns.[13]

5 Discussion & Conclusion

Respiratory Inductance Plethysmography (RIP) was shown to be an effective and safe method of assessing and monitoring respiratory function in people. According to the research examined, RIP has shown to be a particularly valuable evaluation tool for analyzing the physiology of the respiratory system in a variety of settings, both pathological and non-pathological, enabling health practitioners to detect illness at an early stage.

It was discovered that the RIP device's development and enhancement progressed over time. The calibration method and concerns linked to mutual inductance between RIP sensors have been improved as technology has progressed. In terms of mobility and wearability, it was clear that the RIP sensor is now included in the Smart Shirt, making it very helpful in ambulatory settings. In terms of healthcare applications, RIP is widely used. The gadget proved particularly useful in detecting the obstructive sleep apnea-hypopnea syndrome. Additionally, while collecting data from neonates and young patients, the RIP device is useful. Another possible use for the RIP device was in the detection and monitoring of neuromuscular disorders. The pneuRIP was discovered to have a great efficacy in teenagers with neuromuscular disease. The literature research revealed that the design and location of RIP had a significant impact on the system's performance and dependability.

To the best of our knowledge, this is the only review article that gives a short overview of RIP device uses in the medical field. However, the evaluation may be considerably enhanced by including an investigation of the RIP device's accuracy in a larger population and in senior patients. Future work on the design of the RIP device has the potential to be done.

References

1. M. Roesenfeld et al., "An official American Thoracic Society workshop report: optimal lung function tests for monitoring cystic fibrosis, bronchopulmonary dysplasia, and recurrent wheezing in children less than 6 years of age," *Annals of the American Thoracic Society*, vol. 10, no. 2, pp. S1–S11, Apr. 2013, doi: 10.1513/AnnalsATS.201301-017ST.
2. O. H. Mayer, R. G. Clayton, A. F. Jawad, J. M. McDonough, and J. L. Allen, "Respiratory Inductance Plethysmography in Healthy 3- to 5-Year-Old Children *," *Chest*, vol. 124, no. 5, pp. 1812–1819, Nov. 2003, doi: 10.1378/chest.124.5.1812.

3. A. Strang et al., “Measures of respiratory inductance plethysmography (RIP) in children with neuromuscular disease,” *Pediatric Pulmonology*, vol. 53, no. 9, pp. 1260–1268, Jul. 2018, doi: 10.1002/ppul.24134.
4. N. Beydon et al., “An Official American Thoracic Society/European Respiratory Society Statement: Pulmonary Function Testing in Preschool Children,” *American Journal of Respiratory and Critical Care Medicine*, vol. 175, no. 12, pp. 1304–1345, Jun. 2007, doi: 10.1164/rccm.200605-642st.
5. A. A. Hutchison, F. Leclerc, V. Nève, J. J. Pillow, and P. D. Robinson, “The Respiratory System,” *Pediatric and Neonatal Mechanical Ventilation*, pp. 55–112, Apr. 2014, doi: 10.1007/978-3-642-01219-8_4.
6. R. I. Ramos-Garcia, M. H. Imtiaz, E. Sazonov, and S. T. Tiffany, “Evaluation of RIP sensor calibration stability for daily estimation of lung volume,” 2017 Eleventh International Conference on Sensing Technology (ICST), Dec. 2017, doi: 10.1109/icsenst.2017.8304419.
7. K. P. Cohen, D. Panescu, J. H. Booske, J. G. Webster, and W. J. Tompkins, “Design of an inductive plethysmograph for ventilation measurement,” *Physiological Measurement*, vol. 15, no. 2, pp. 217–229, May 1994, doi: 10.1088/0967-3334/15/2/009.
8. Z. Zhang, J. Zheng, H. Wu, W. Wang, B. Wang, and H. Liu, “Development of a Respiratory Inductive Plethysmography Module Supporting Multiple Sensors for Wearable Systems,” *Sensors*, vol. 12, no. 10, pp. 13167–13184, Sep. 2012, doi: 10.3390/s121013167.
9. A. M. Sinton and R. Suntheralingam, “Respiratory inductance plethysmography with an electrical impedance plethysmograph,” *Medical & Biological Engineering & Computing*, vol. 26, no. 2, pp. 213–217, Mar. 1988, doi: 10.1007/bf02442267.
10. Zhang, Z., Mengsun, Y. U., Ruoxin, L. I., Taihu, W. U., & Jialong, W. U. (2006). Design of a wearable respiratory inductive plethysmograph and its applications. *Space Medicine & Medical Engineering*, (05).
11. J. W. Zheng, Z. B. Zhang, T. H. Wu, and Y. Zhang, “A wearable mobihealth care system supporting real-time diagnosis and alarm,” *Medical & Biological Engineering & Computing*, vol. 45, no. 9, pp. 877–885, Jul. 2007, doi: 10.1007/s11517-007-0221-y.
12. T. Rahman, R. Page, C. Page, J.-R. Bonnefoy, T. Cox, and T. H. Shaffer, “pneuRIPTM: A Novel Respiratory Inductance Plethysmography Monitor,” *Journal of Medical Devices*, vol. 11, no. 1, Jan. 2017, doi: 10.1115/1.4035546.

13. M. B. Phillips, J. Beach, M. Cathey, J. Lockert, and W. Satterfield, "Reliability and Validity of the Hexoskin Wearable Body Metrics Telemetry Shirt", *J Sport Human Perf*, vol. 5, no. 2, Sep. 2017.
14. Cherif, N.H.; Mezghani, N.; Gaudreault, N.; Ouakrim, Y.; Mouzoune, I.; Boulay, P. Physiological data validation of the hexoskin smart textile. In *Proceedings of the 11th International Joint Conference on Biomedical Engineering Systems and Technologies (BIOSTEC 2018)*, Funchal, Portugal, 19–21 January 2018; Volume 1, pp. 150–156.
15. P. Lopez-Meyer, S. Tiffany, Y. Patil, and E. Sazonov, "Monitoring of Cigarette Smoking Using Wearable Sensors and Support Vector Machines," *IEEE Transactions on Biomedical Engineering*, vol. 60, no. 7, pp. 1867–1872, Jul. 2013, doi: 10.1109/tbme.2013.2243729.
16. S. M. Sale, "Neonatal apnoea," *Best Practice & Research Clinical Anaesthesiology*, vol. 24, no. 3, pp. 323–336, Sep. 2010, doi: 10.1016/j.bpa.2010.04.002.
17. M. P. Mansukhani, B. P. Kolla, and K. Ramar, "International Classification of Sleep Disorders 2 and American Academy of Sleep Medicine Practice Parameters for Central Sleep Apnea," *Sleep Medicine Clinics*, vol. 9, no. 1, pp. 1–11, Mar. 2014, doi: 10.1016/j.jsmc.2013.10.006.
18. A. Sabil, C. Marien, M. LeVaillant, G. Baffet, N. Meslier, and F. Gagnadoux, "Diagnosis of sleep apnea without sensors on the patient's face," *Journal of Clinical Sleep Medicine*, vol. 16, no. 7, pp. 1161–1169, Jul. 2020, doi: 10.5664/jcsm.8460.
19. K. A. Brown, A. A. Aoude, H. L. Galiana, and R. E. Kearney, "Automated respiratory inductive plethysmography to evaluate breathing in infants at risk for postoperative apnea," *Canadian Journal of Anesthesia/Journal canadien d'anesthésie*, vol. 55, no. 11, pp. 739–747, Nov. 2008, doi: 10.1007/bf03016346.
20. C. D. Kurth and S. E. LeBard, "Association of Postoperative Apnea, Airway Obstruction, and Hypoxemia in Former Premature Infants," *Anesthesiology*, vol. 75, no. 1, pp. 22–26, Jul. 1991, doi: 10.1097/00000542-199107000-00005.
21. S. Malviya, J. Swartx, and J. Lerman, "Are All Preterm Infants Younger than 60 Weeks Postconceptual Age at Risk for Postanesthetic Apnea?" *Anesthesiology*, vol. 78, no. 6, pp. 1076–1081, Jun. 1993, doi: 10.1097/00000542-199306000-00009.

22. A. J. Davidson et al., “Apnea after Awake Regional and General Anesthesia in Infants,” *Anesthesiology*, vol. 123, no. 1, pp. 38–54, Jul. 2015, doi: 10.1097/aln.0000000000000709.
23. C. A. Robles-Rubio, R. E. Kearney, G. Bertolizio, and K. A. Brown, “Automatic unsupervised respiratory analysis of infant respiratory inductance plethysmography signals,” *PLOS ONE*, vol. 15, no. 9, p. e0238402, Sep. 2020, doi: 10.1371/journal.pone.0238402.
24. C. Landon, “Novel Methods of Ambulatory Physiologic Monitoring in Patients with Neuromuscular Disease,” *Pediatrics*, vol. 123, no. Supplement 4, pp. S250–S252, May 2009, doi: 10.1542/peds.2008-2952l.
25. A. Strang et al., “Measures of respiratory inductance plethysmography (RIP) in children with neuromuscular disease,” *Pediatric Pulmonology*, vol. 53, no. 9, pp. 1260–1268, Jul. 2018, doi: 10.1002/ppul.24134.
26. H. T. Ngo, C. V. Nguyen, T. M. H. Nguyen, and T. Van Vo, “A Portable Respiratory Monitor Using Respiratory Inductive Plethysmography,” *IFMBE Proceedings*, pp. 222–225, 2013, doi: 10.1007/978-3-642-32183-2_57.